

JAEHOON HAHM

Updated: October 14, 2025

Email: jh141@illinois.edu Github: github.com/jaehoon-hahm Homepage: jaehoon-hahm.github.io

RESEARCH INTEREST

Generative AI + Science, ML + Quantum Many-body Physics, Geometric Diffusion/
Flow/ Markov-Matching Models, Neural Quantum States.

EDUCATION

University of Illinois Urbana-Champaign (UIUC)

Ph.D., Physics

2024.08 – Current

Advisor: Bryan K. Clark

Research: Machine Learning for Quantum Many-body Physics, ML + Material Science,
Computational Condensed Matter

Graduate School of Data Science, Seoul National University (SNU)

M.S., Data Science

2022.03 – 2024.02

- Thesis: *Isometric Representation Learning for Disentangled Latent Space of Diffusion Models*

- Advisor: Joonseok Lee

Seoul National University (SNU)

B.S., Physics, minor in Mathematics

2016.03 – 2022.02

- Thesis: *Improvement in Variational Quantum Algorithms by Measurement Simplification*

- Advisor: Young June Park

- Mandatory Military Service: Republic of Korea Army, 2019.07 – 2021.02

Seoul Science High School

Graduated with Distinction

2013.03 – 2016.02

- 1 year early admission to a science gifted high school

PUBLICATIONS

[P3] “Equivariant Latent Alignment via Flow Matching under Group Symmetries”
Sunghyun Kim*, [Jaehoon Hahm](#)*, Jeongwoo Shin, Joonseok Lee[†]
Under Review at ICLR 2025

[P2] “Shadow Flow Matching for Learning Quantum States”
[Jaehoon Hahm](#), Bryan Clark[†]
Under Review at ICLR 2025

[C3] “SummDiff: Generative Modeling of Video Summarization with Diffusion”
Kwanseok Kim*, [Jaehoon Hahm](#)*, Sumin Kim, Jinhwan Sul, Byunghak Kim, Joonseok Lee[†]
International Conference on Computer Vision (ICCV) 2025

[C2] “Generative Modeling of Quantum Distribution with Functional Flow Matching”
[Jaehoon Hahm](#)*, Tak Hur*, Joonseok Lee, Daniel K. Park[†]
Quantum Techniques in Machine Learning (QTML) 2024

[C1] “Isometric Representation Learning for Disentangled Latent Space of Diffusion Models”
[Jaehoon Hahm](#)*, Junho Lee*, Sunghyun Kim, Joonseok Lee[†]
International Conference on Machine Learning (ICML) 2024

[P1] “Improvement in Variational Quantum Algorithms by Measurement Simplification”
[Jaehoon Hahm](#), Hayeon Kim, Young June Park
 arXiv:2312.06176

WORK EXPERIENCE **Undergraduate Research Assistant** 2018.07 – 2019.04, 2021.02 – 2021.12
 Advisor: Young June Park, *Samsung Beyond Limit* project ECE, SNU
 - Proposed a “Measurement Simplification” pre-processing method that accelerates variational quantum circuit computation.
 - Experiments of Quantum KNN on IBM Qiskit quantum computers.

Undergraduate Research Assistant 2021.02 – 2021.12
 Advisor: Taehyun Kim CSE, SNU
 - Measured gate fidelities of the group’s trapped-ion quantum computer using randomized benchmarking protocol, by self-developed control framework based on IBM Qiskit.

Undergraduate Research Assistant 2021.07 – 2021.12
 Advisor: Hyunyoung Choi Physics, SNU
 - Performed diamond NV center control experiments: Rabi oscillation, spin echo, and dynamical decoupling.

Republic of Korea Army 2019.07 – 2021.02
Chemical, Biological, Radiological, and Nuclear Reconnaissance, 5th Infantry Division
 - 1 year and 7 months of mandatory military service

PATENT **Korea Patent**
 SummDiff: Generative Modeling of Video Summarization with Diffusion 2025

TEACHING EXPERIENCE **TA, UIUC**
 Phys212: University Physics: Electricity & Magnetism 2025
 Phys435: Electromagnetic Fields I 2024

LANGUAGES **Programming:** Python, C, C++, Java, MATLAB, Mathematica
Frameworks: PyTorch, JAX, TensorFlow, NumPy, Pandas
Quantum: IBM Qiskit, Google Cirq, PennyLane, TensorFlow Quantum
Natural Languages: English (Professional), Korean (Native)

REFERENCES **Bryan K. Clark** 2024.08 - Current
 Ph.D. Advisor (UIUC) bkclark@illinois.edu
Joonseok Lee 2022.03 - 2024.02
 M.S. Advisor (SNU) joonsek@snu.edu
Rainer Engelken 2025.09 - Current
 Instructor (UIUC) engelken@illinois.edu